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EXPERIMENT 19

Edge Detection using the Sobel Operator along both X and Y Axes

AIM

To perform edge detection on the given image by combining the Sobel gradients along the X and Y axes.

PROGRAM

```
import cv2
import numpy as np

img = cv2.imread("image1.jpg", cv2.IMREAD_GRAYSCALE)

sobel_x = cv2.Sobel(img, cv2.CV_64F, 1, 0, ksize=3)
sobel_y = cv2.Sobel(img, cv2.CV_64F, 0, 1, ksize=3)

# Gradient magnitude: sqrt(Gx^2 + Gy^2)
magnitude = cv2.magnitude(sobel_x, sobel_y)
edges = cv2.convertScaleAbs(magnitude)

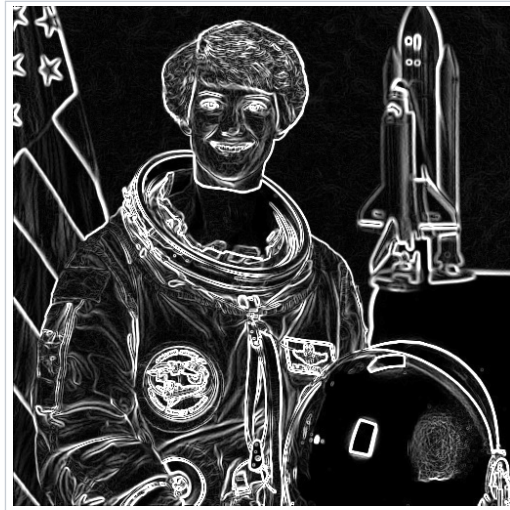
cv2.imwrite("sobel_xy.jpg", edges)
cv2.imshow("Original", img)
cv2.imshow("Sobel XY", edges)
cv2.waitKey(0)
cv2.destroyAllWindows()
```

INPUT



image1.jpg (input image)

OUTPUT



sobel_xy.jpg (gradient magnitude combining the X and Y edges)

RESULT

Thus the edges along both the X and Y axes were successfully detected and combined using the Sobel operator.